

Ocean Acidification and other Climate Change Implications

**A deeper look at the effects of growing CO₂ and climate
change on the ocean**

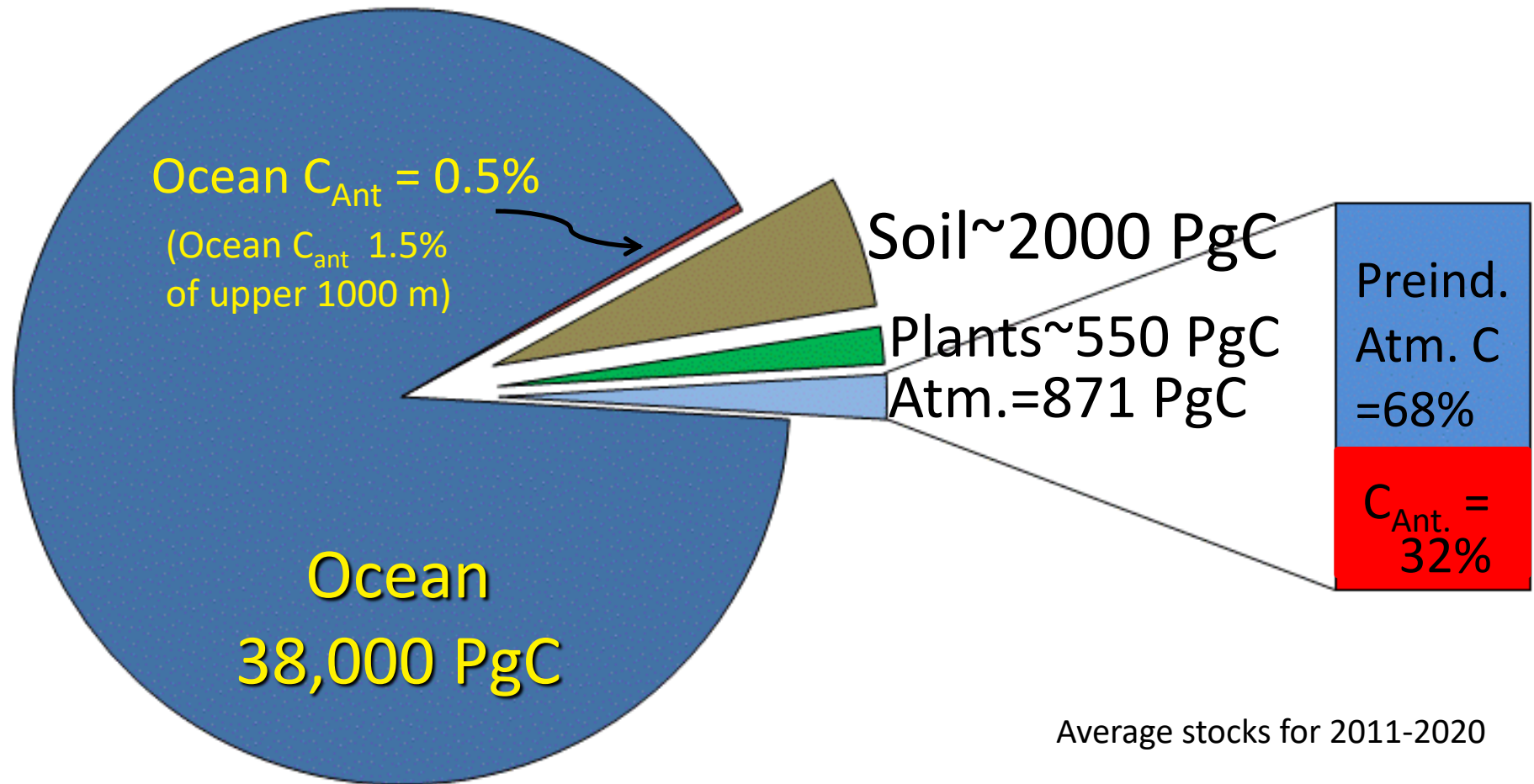
OCN 623 – Chemical Oceanography

Student Learning Outcomes (SLOs)

At the completion of today's section, students should be able to:

1. Know the basic definition of Revelle Factor and how it will change with rising CO₂
2. Describe the impact of rising CO₂ on calcifying organisms
3. List some of the concerns for OA on marine ecosystems
4. Understand how IPCC views risk for climate change and OA

Carbon Inventories of Reservoirs that Naturally Exchange Carbon on Time Scales of Decades to Centuries



- Oceans contain ~90% of carbon in this 4 component system
- anthropogenic component is difficult to detect in the ocean

Ocean Carbon Chemistry Review

$\text{CO}_{2(\text{gas})}$



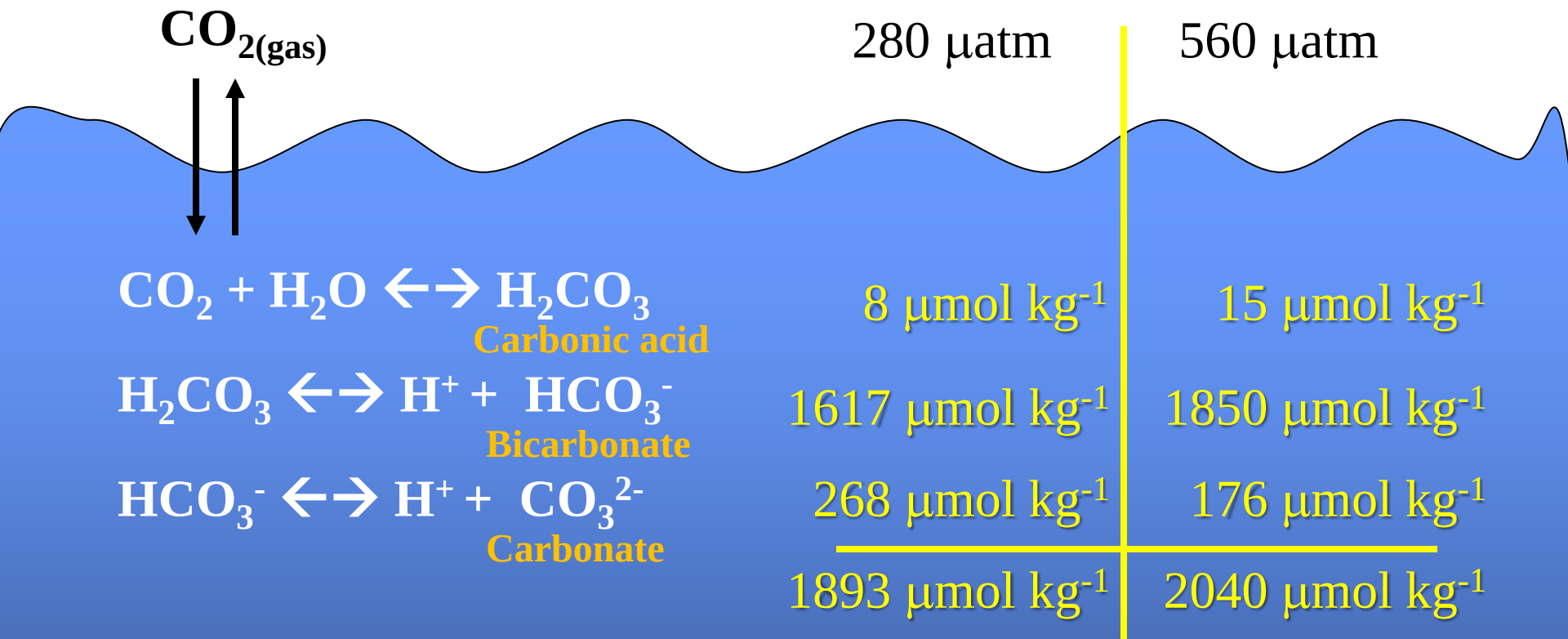
$$p\text{CO}_2 = x\text{CO}_2 \cdot P$$

$$\text{TCO}_2 = [\text{CO}_2^*] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$$

$$\text{TA} = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + [\text{B}(\text{OH})_4^-] + \dots - [\text{H}^+]$$

$$\text{pH} = -\log\{\text{H}^+\}$$

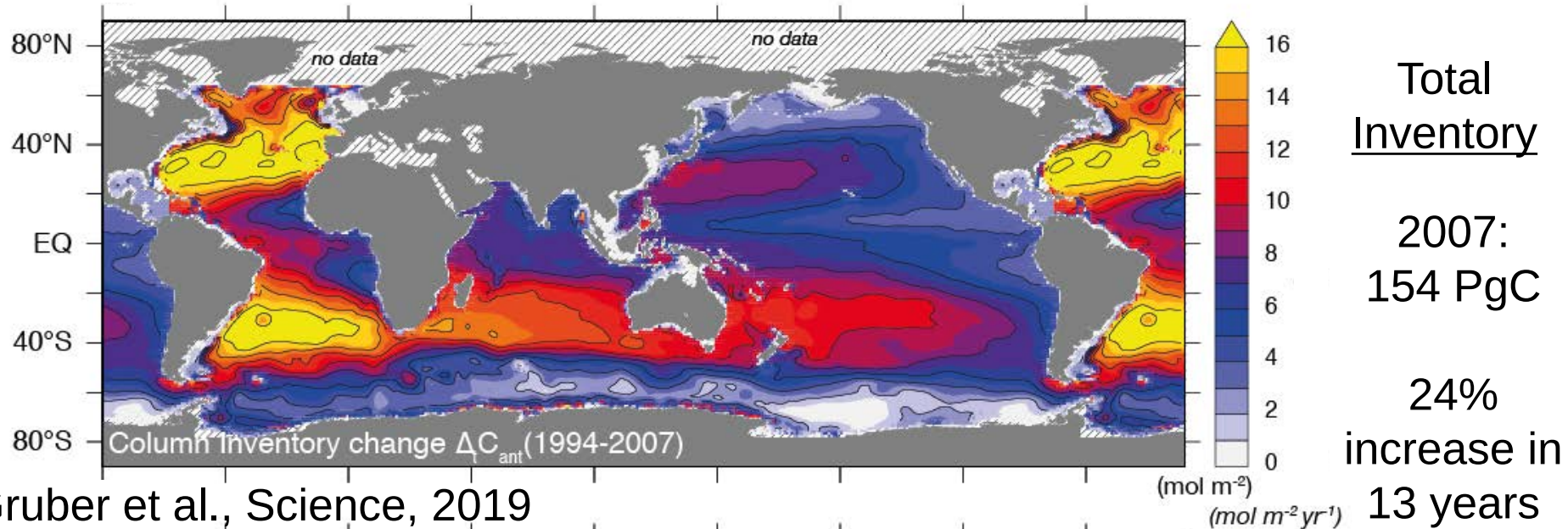
Ocean Carbon Chemistry Review



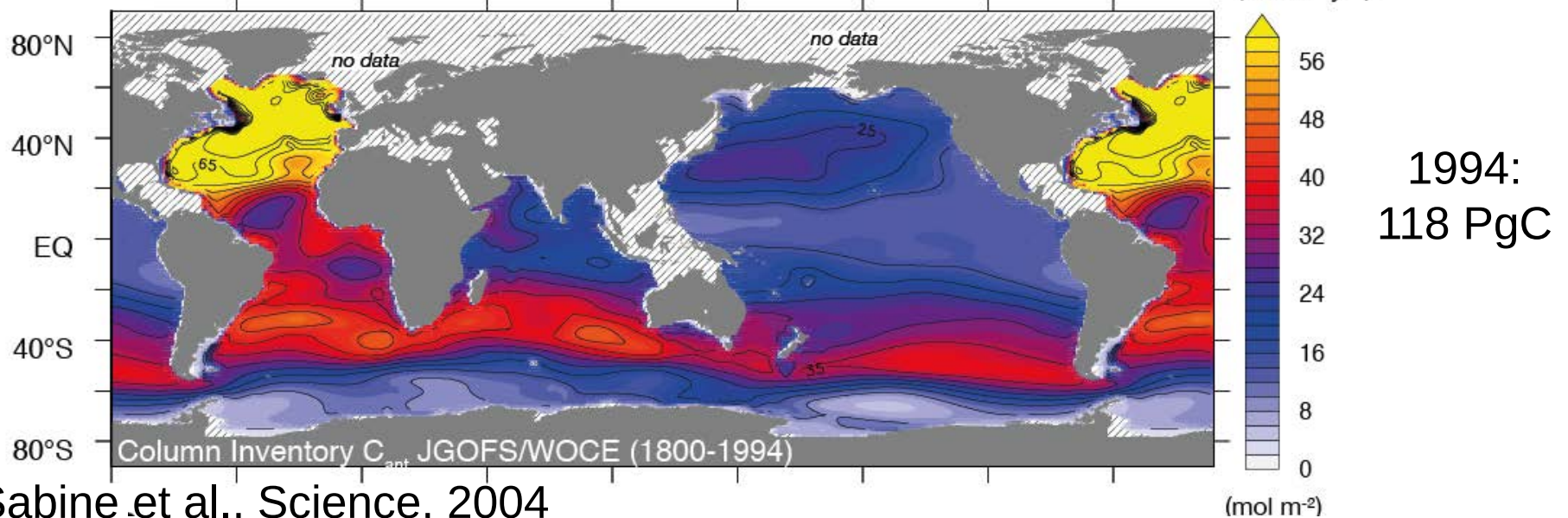
100% $\Delta p\text{CO}_2 \rightarrow 8\% \Delta \text{TCO}_2$

Taken from Feely et al. (2001)

Measurement-Based Inventory Change Estimates Confirm An Average Global Uptake Rate of 2.6 ± 0.3 PgC/yr between 1994 and 2007

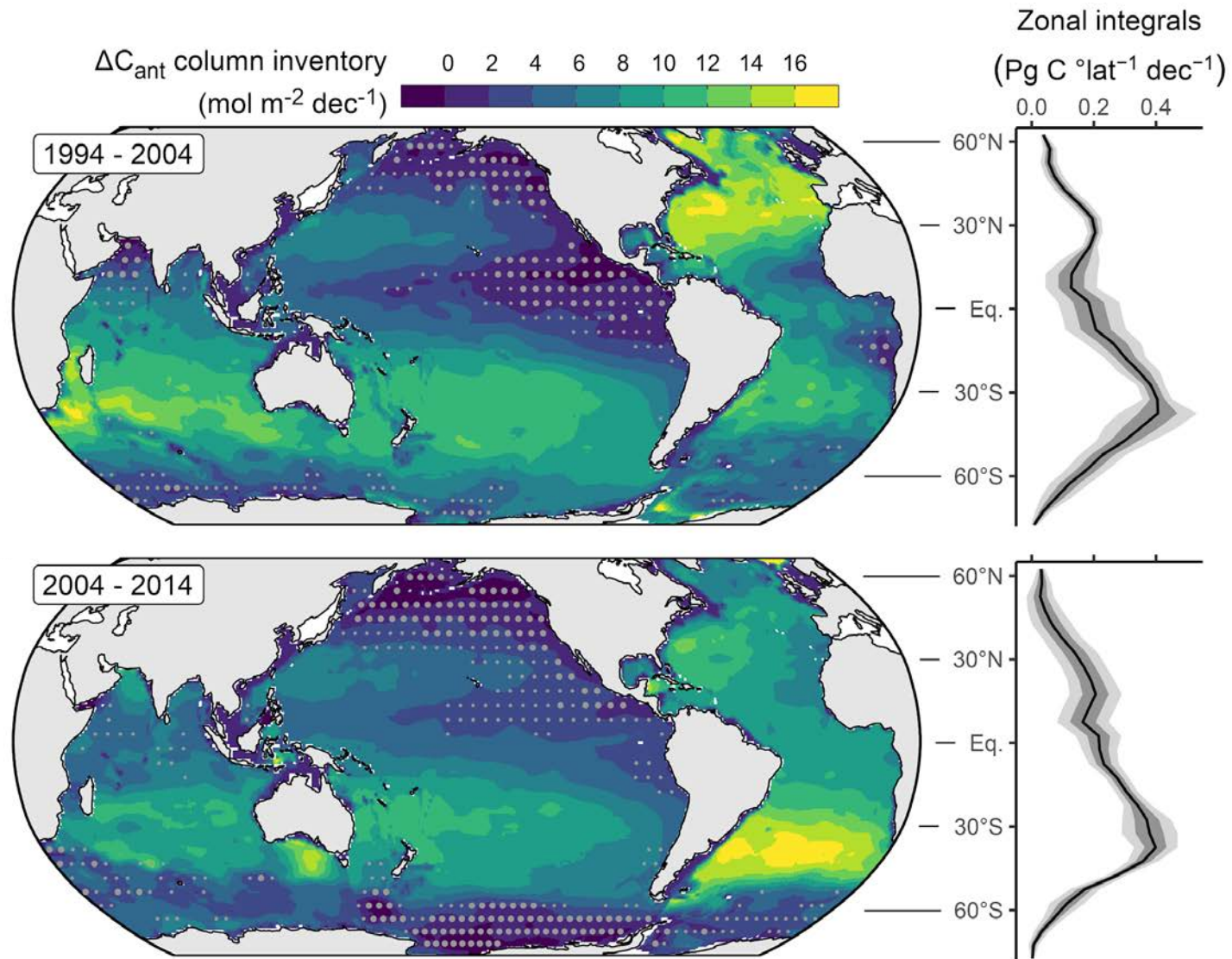


Gruber et al., Science, 2019



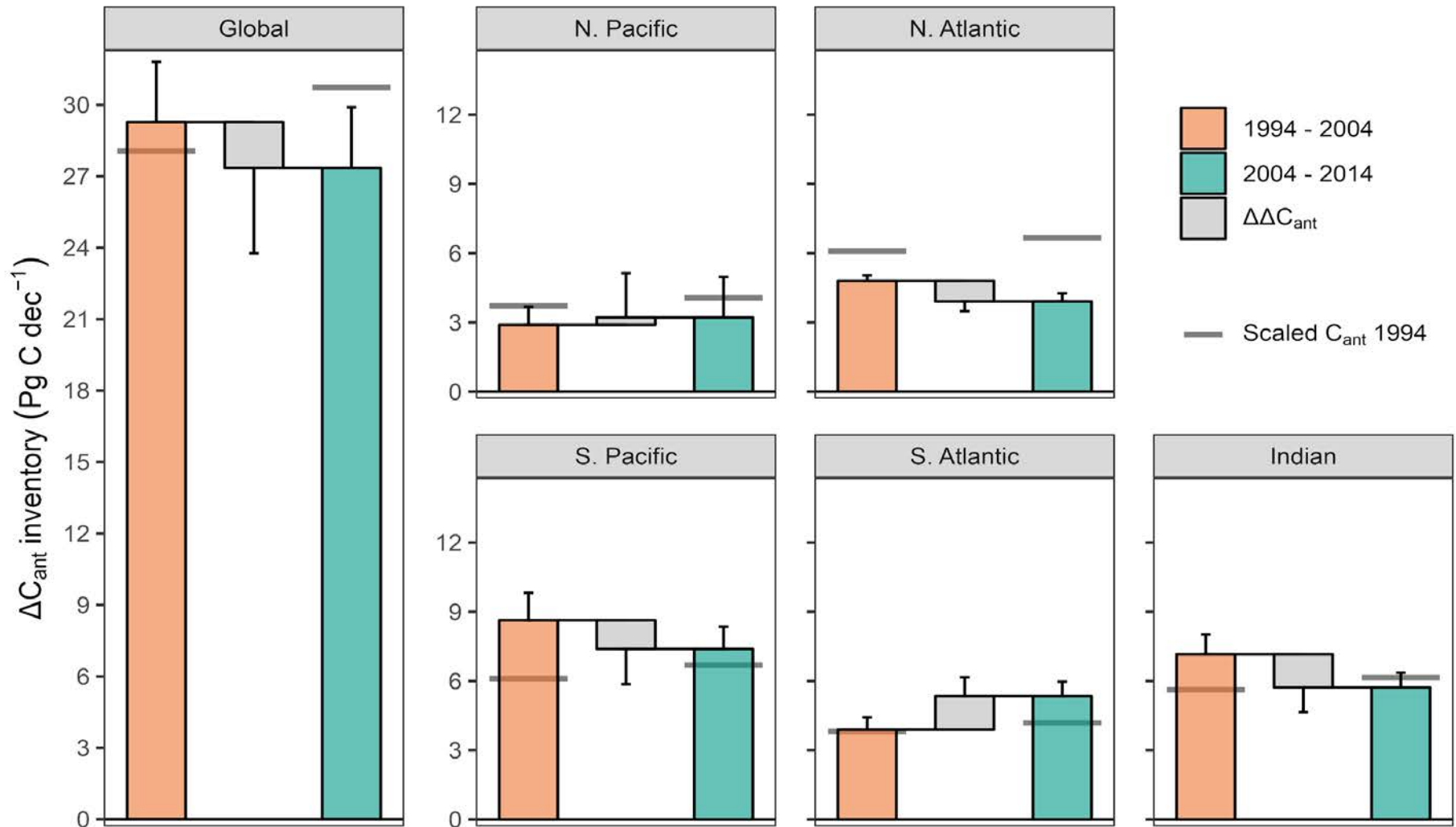
Sabine et al., Science, 2004

Column Inventory Maps and Zonal Integrals of Anthropogenic Carbon



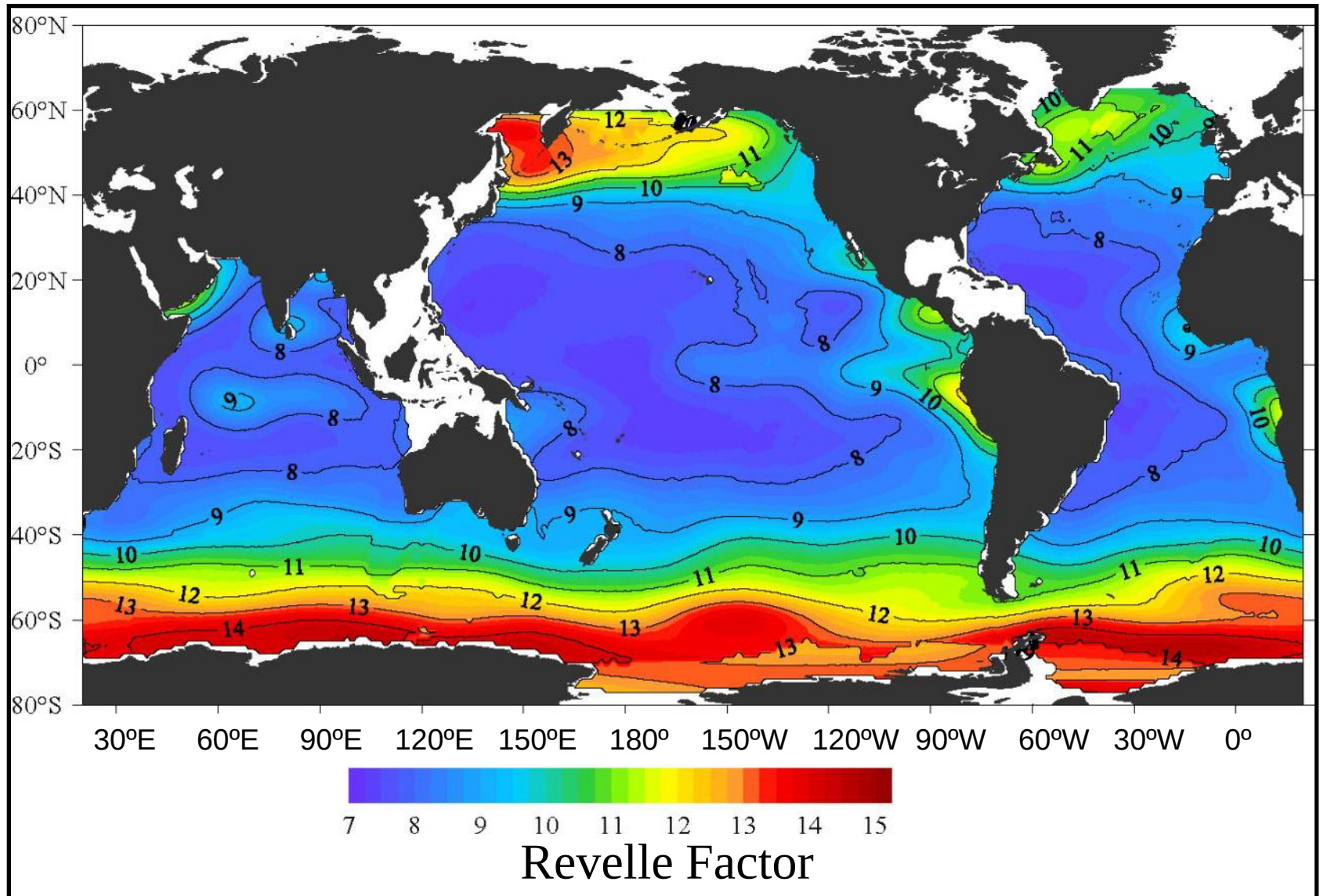
Müller et al. (2023)

Change in Anthropogenic Carbon Storage Over Two Decades



Müller et al. (2023)

Global Distribution of Surface Revelle Factor



R. Revelle, H., E. Suess, *Tellus* 9, 18 (1957)

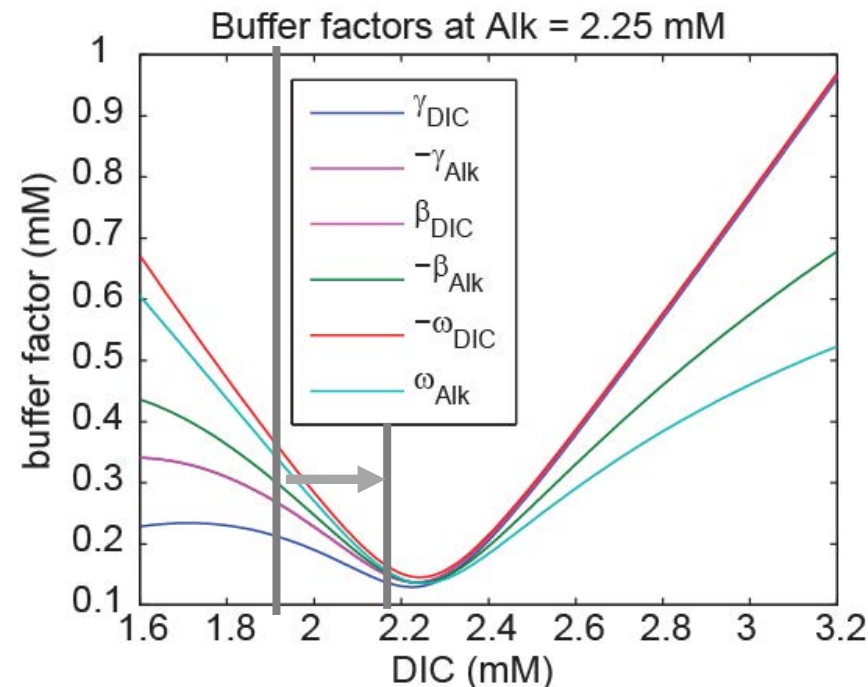
$$\frac{(\Delta f\text{CO}_2 / \Delta \text{DIC})}{(f\text{CO}_2 / \text{DIC})}$$

Egleston et al. (GBC, 2010) define six new buffer factors, each of which can be explicitly calculated:

$$\gamma_{\text{DIC}} = \left(\frac{\partial \ln[\text{CO}_2]}{\partial \text{DIC}} \right)^{-1} = \text{DIC} - \frac{\text{Alk}_c^2}{S} \quad S = [\text{HCO}_3^-] + 4[\text{CO}_3^{2-}] + \frac{[\text{H}^+][\text{B(OH)}_4^-]}{K_{hb} + [\text{H}^+]} + [\text{H}^+] - [\text{OH}^-]$$

Revelle Factor = $\text{DIC} / \gamma_{\text{DIC}}$

equivalent to traditional Revelle Factor that can be used to determine the efficiency of the ocean sink as CO_2 rises

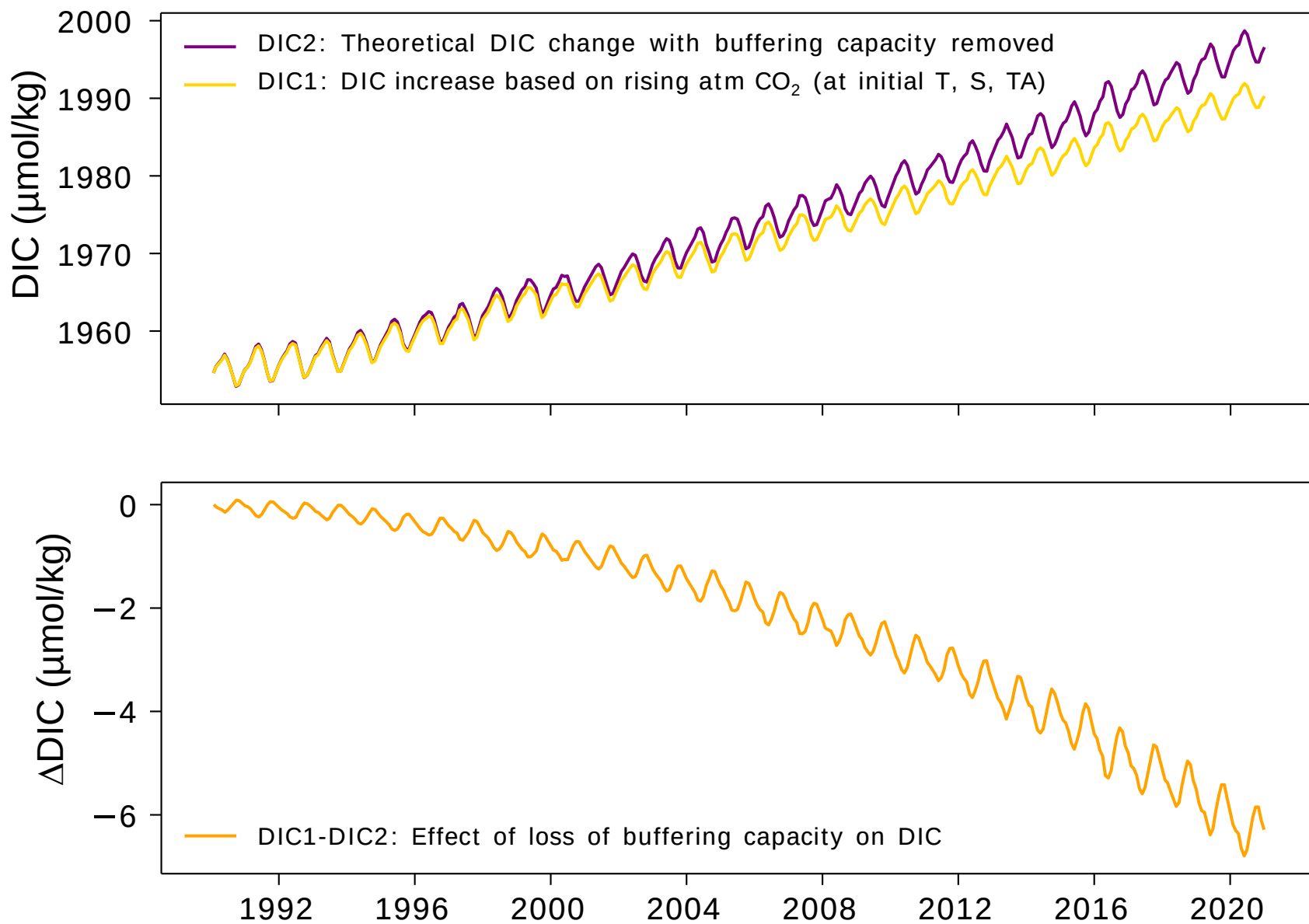


Between the preindustrial and 2000 we expect to see a 25% decrease in uptake efficiency

A doubling of CO_2 (560 ppm) results in a 60% decrease in uptake efficiency

A tripling of CO_2 (840 ppm) results in a 75% decrease in uptake efficiency

Effect of Buffering Capacity on DIC at Station ALOHA



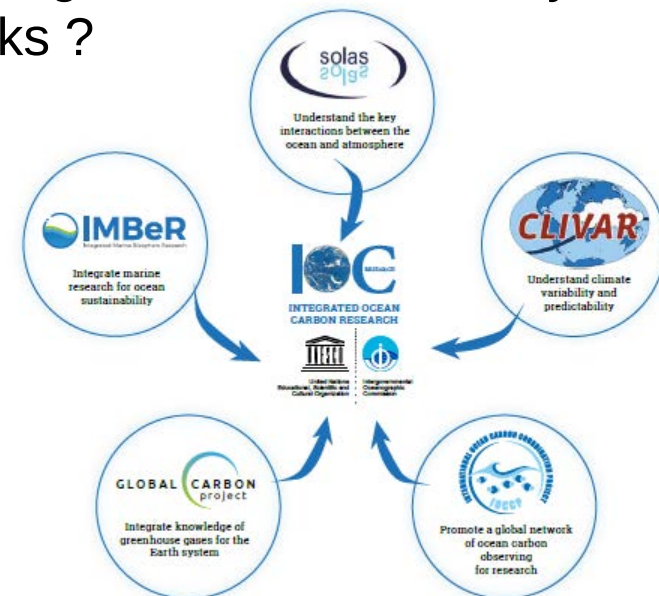
Fundamental and Emerging Research Questions

Will the ocean uptake of anthropogenic carbon dioxide continue as primarily an abiotic process ?

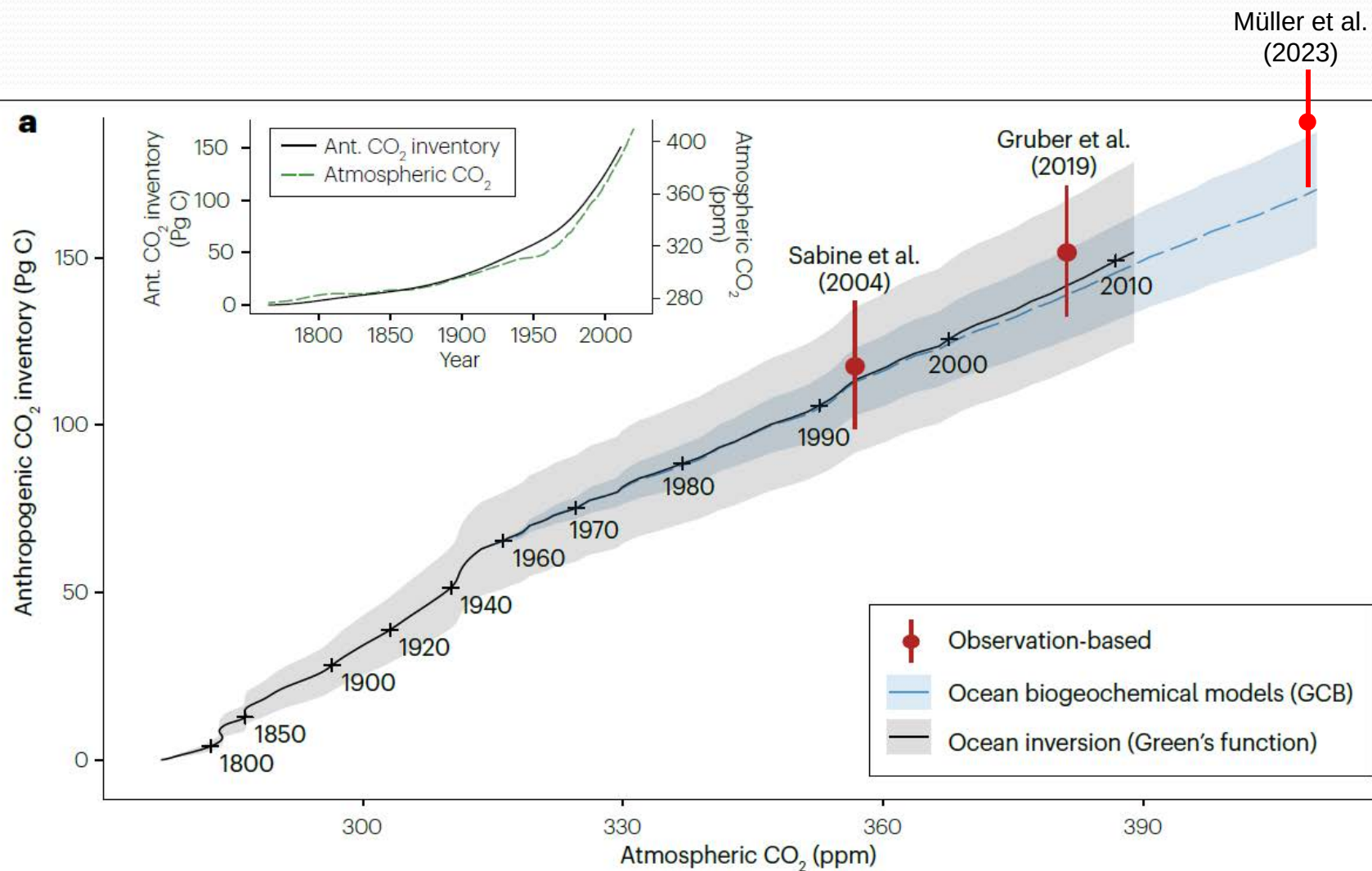
What is the role of biology in the ocean carbon cycle and how is it changing ?

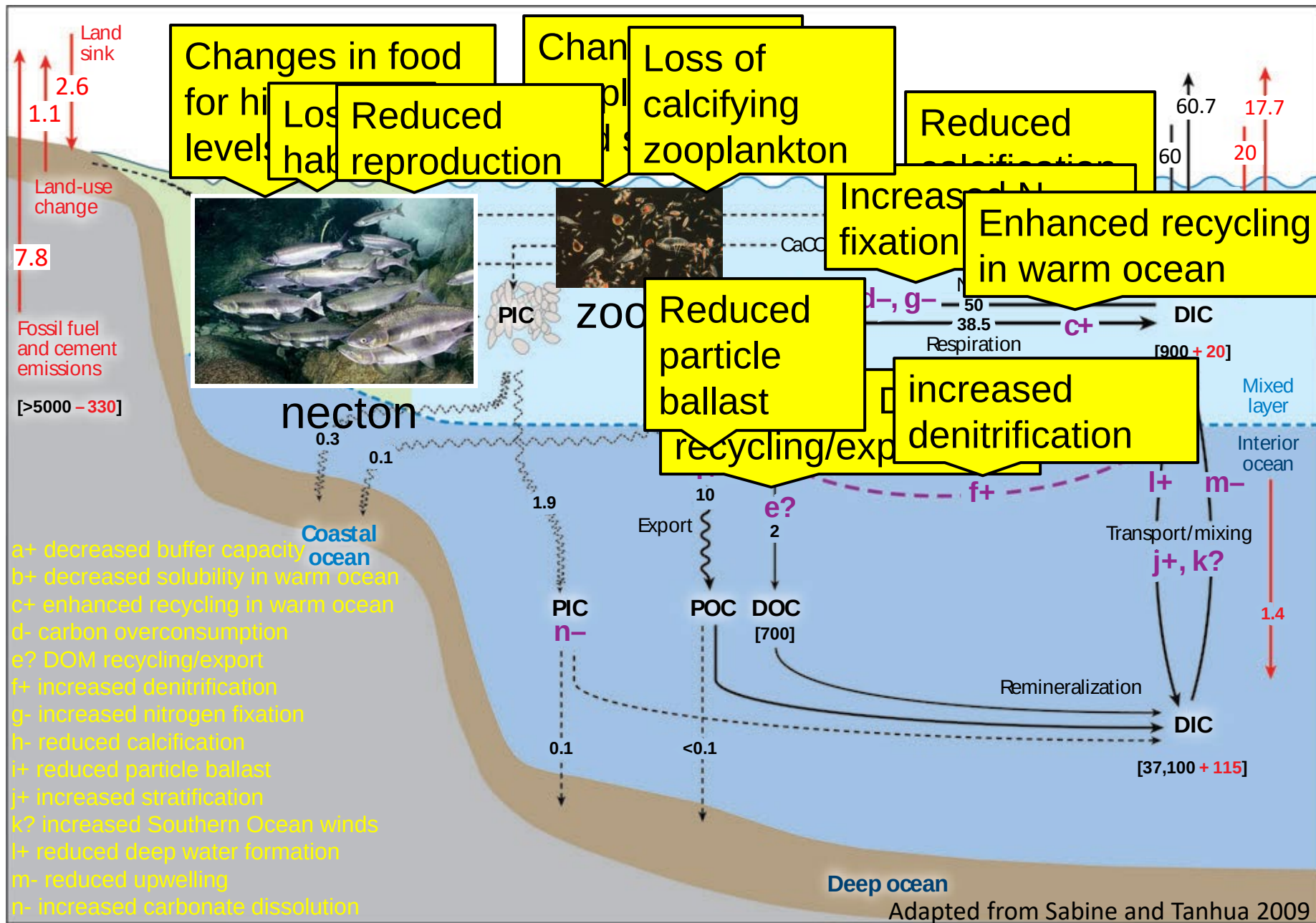
What are the exchanges of carbon between the land-ocean-ice continuum ?

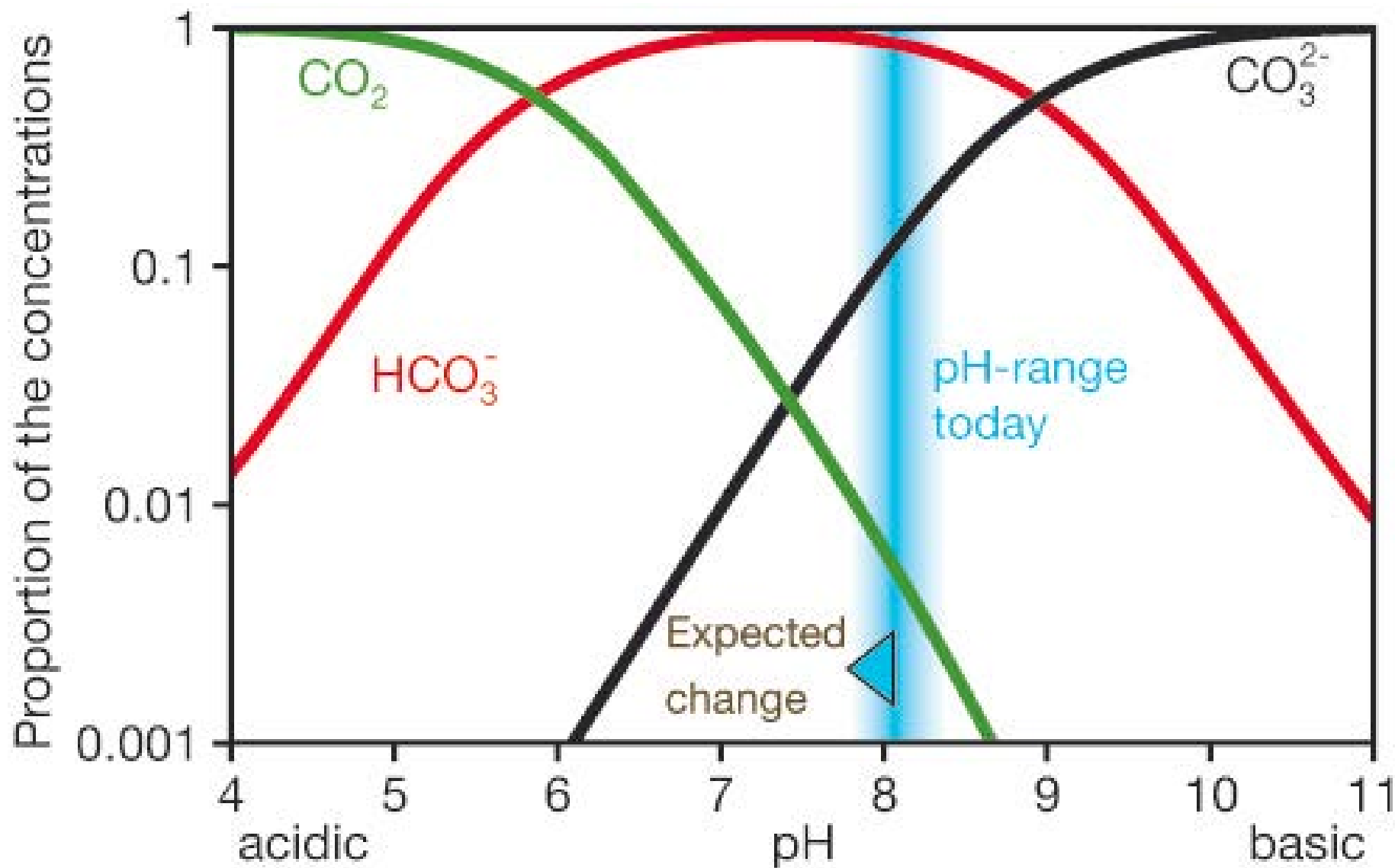
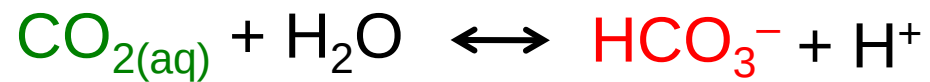
How are humans altering the ocean carbon cycle and resulting feedbacks ?



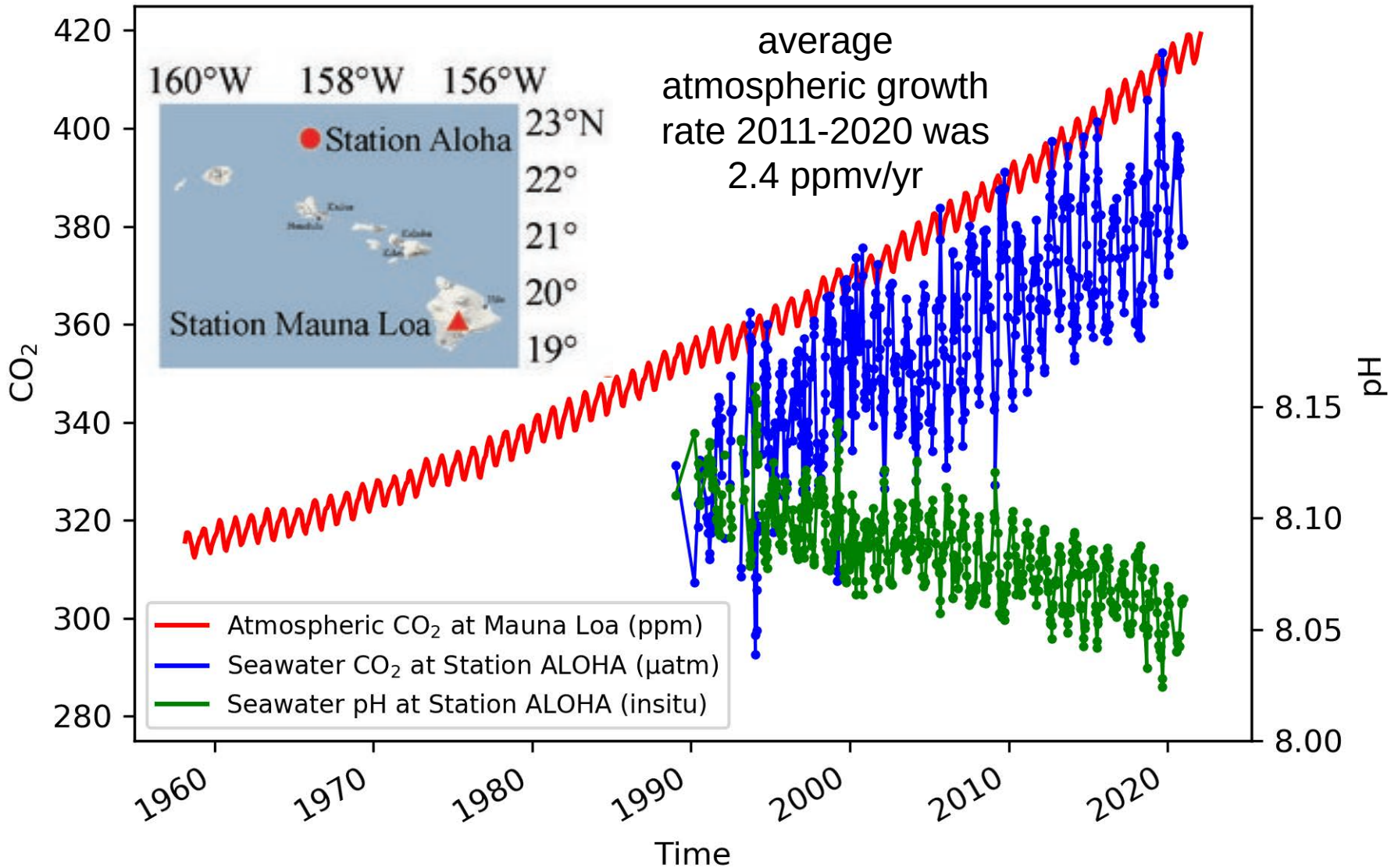
Anthropogenic CO₂ Inventory Increase Follows Atmospheric Trend





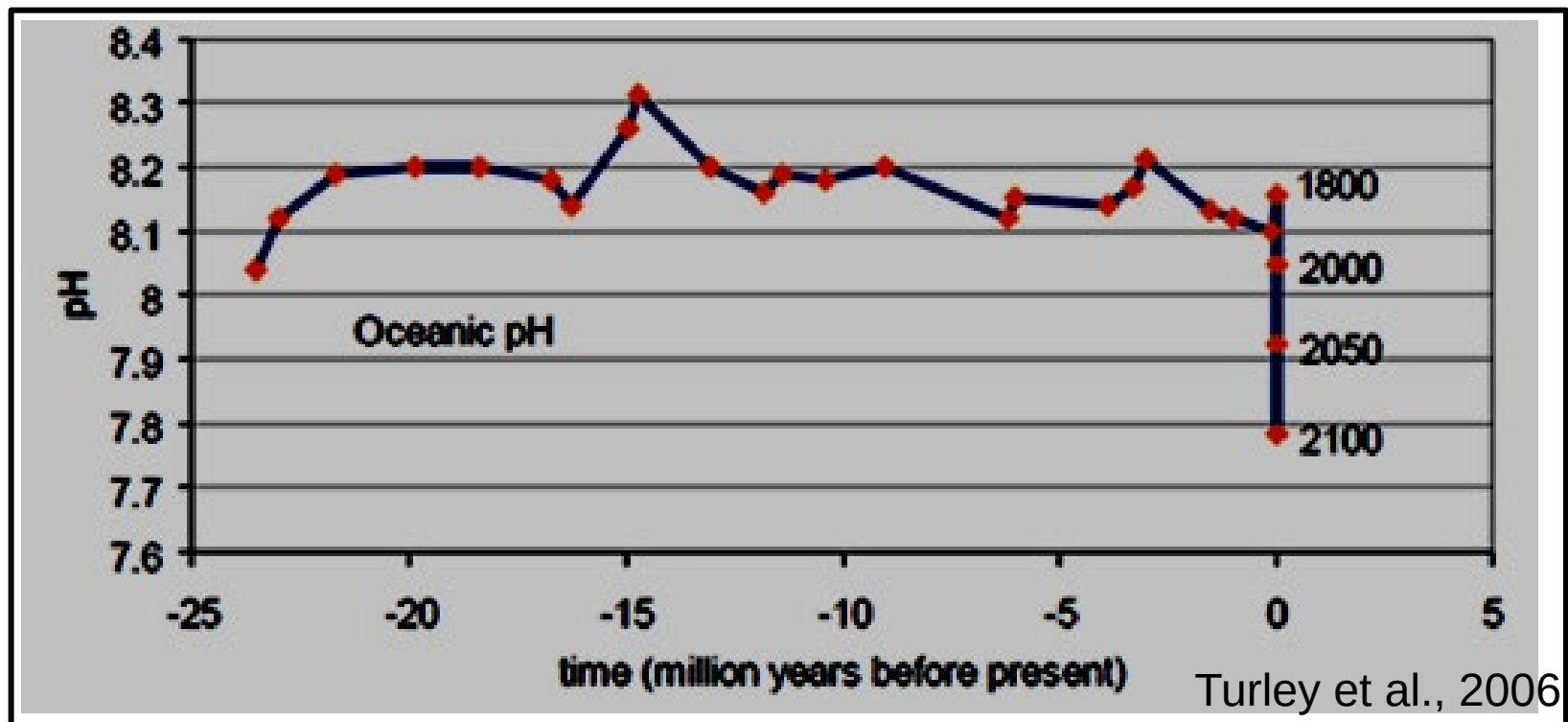


Surface Ocean pH Decreases as CO₂ Increases

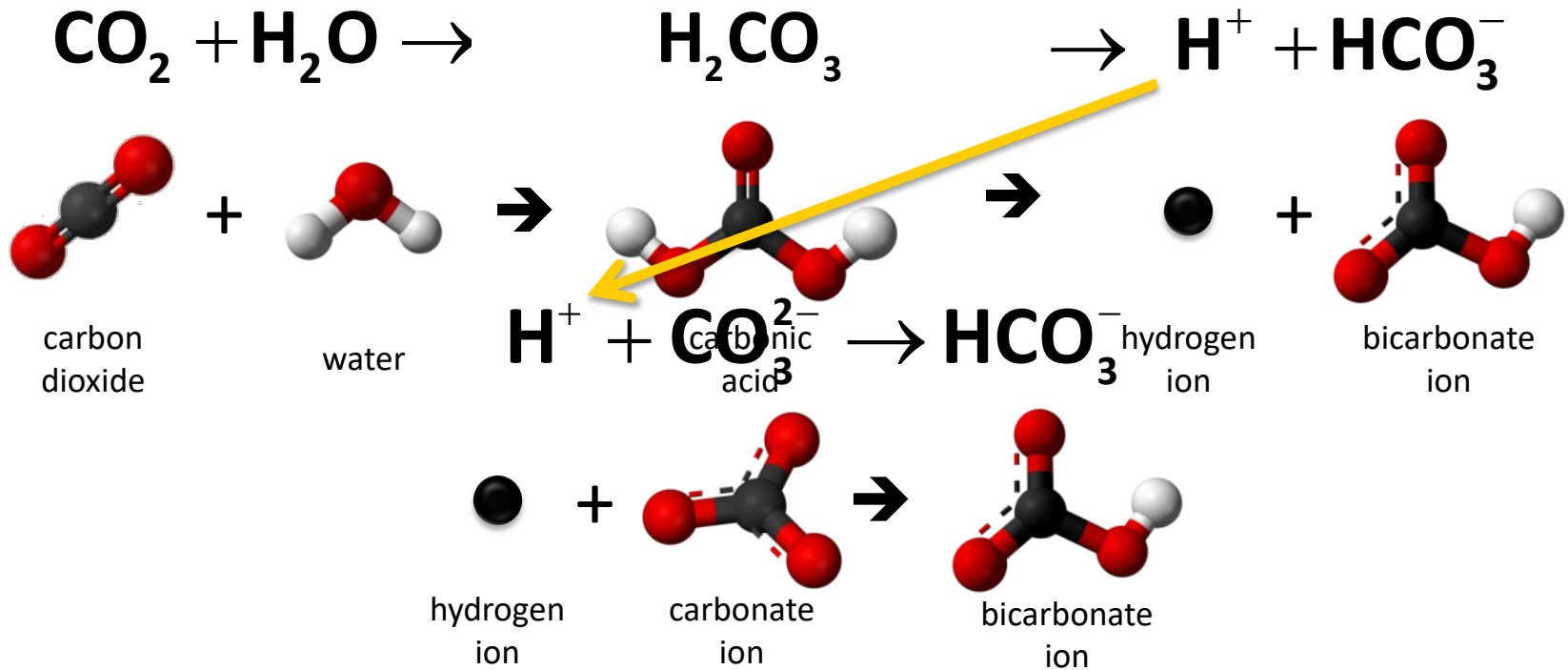


Ocean Acidification: Historical Perspective

- For the last 20 Million years the pH of the ocean has remained relatively stable between approximately 8.1 and 8.2
- The uptake of anthropogenic CO_2 has lowered ocean pH by 0.1, representing a 28% increase in acidity over the last 200 years.
- The estimated drop in pH by the end of the century is not only larger than seen over the last 20 million years, but is also at least 100 times faster than in the past.

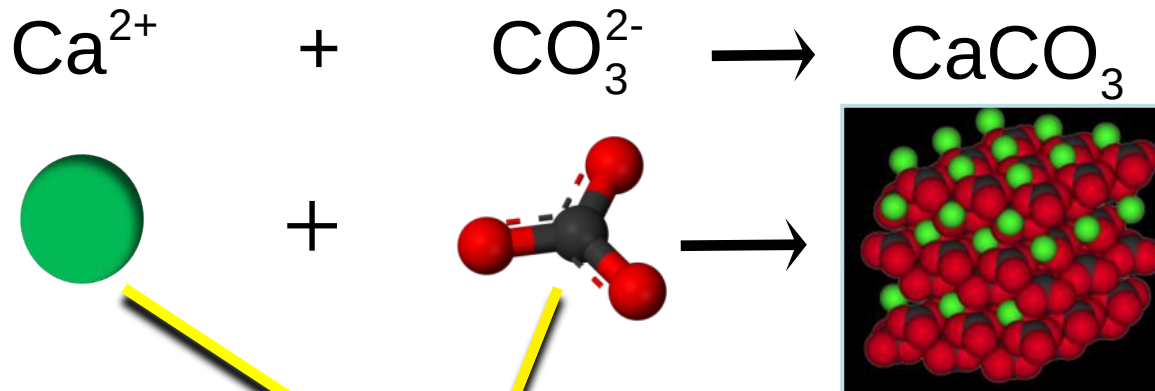


Ocean Acidification: Fundamental Chemistry



- More than 99% of the H⁺ formed consume CO₃²⁻ to form HCO₃⁻ making it more difficult for organisms to form their shells.

Ocean Acidification: Fundamental Chemistry



Calcification Index

Saturation State

$$\Omega_{\text{phase}} = \frac{[\text{Ca}^{2+}][\text{CO}_3^{2-}]}{K_{\text{sp,phase}}^*}$$

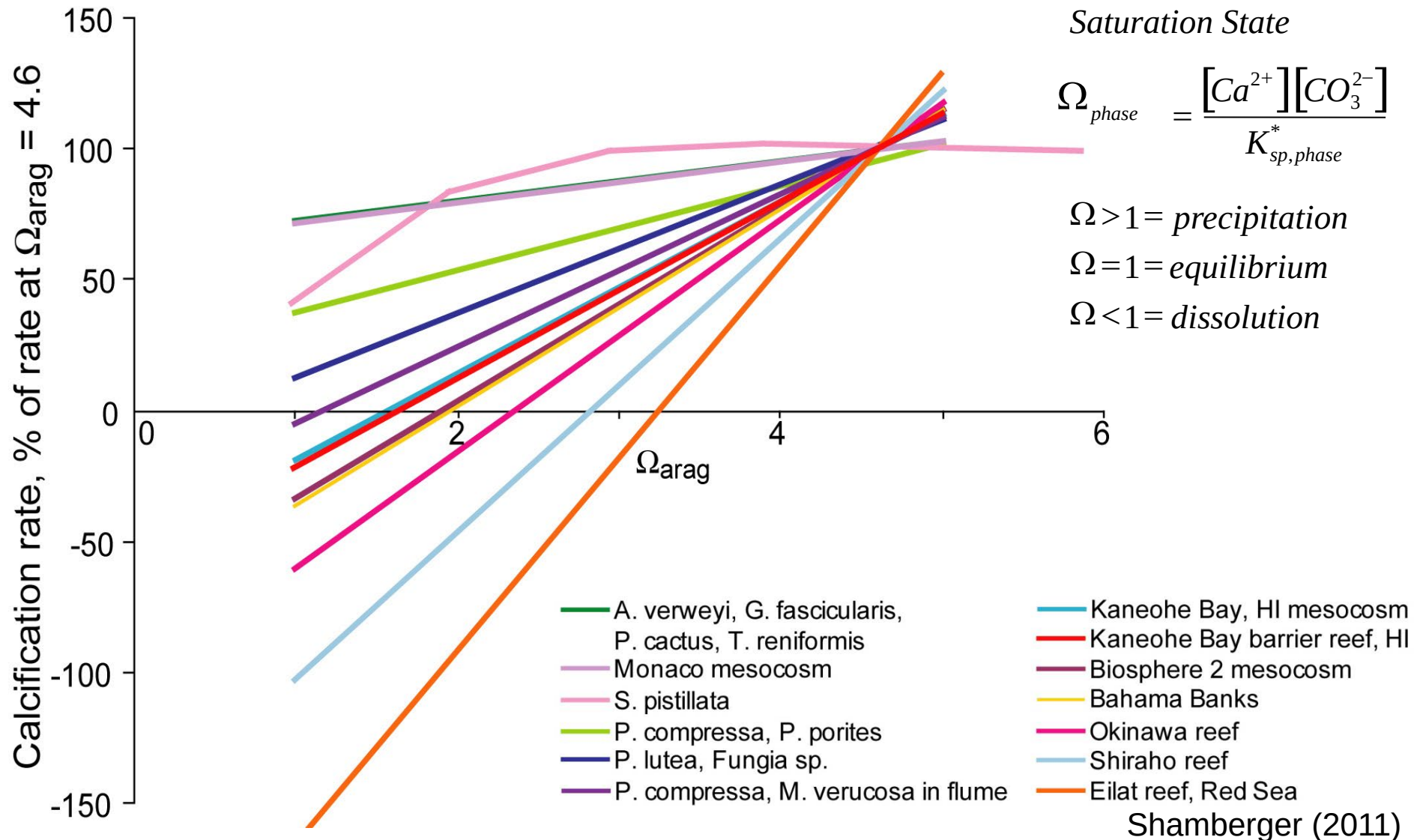
$\Omega > 1$ CaCO_3 stable

$\Omega = 1$ equilibrium

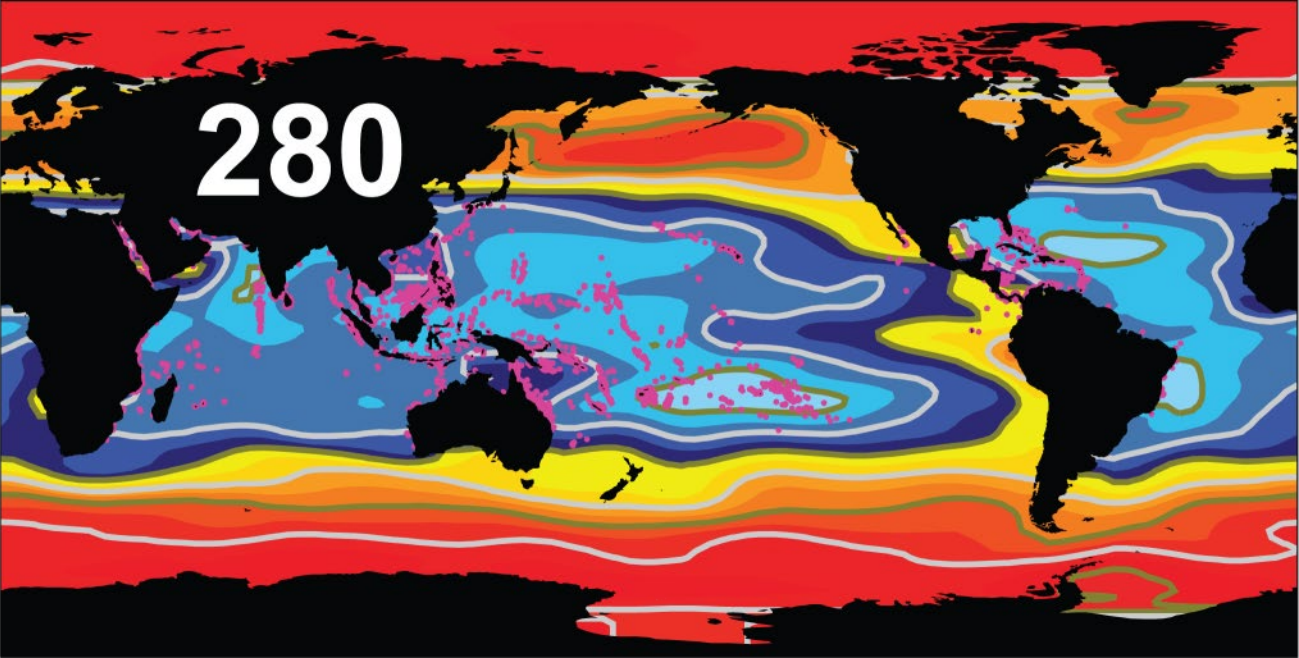
$\Omega < 1$ CaCO_3 dissolves



Corals show a strong response to high CO₂/ low saturation state

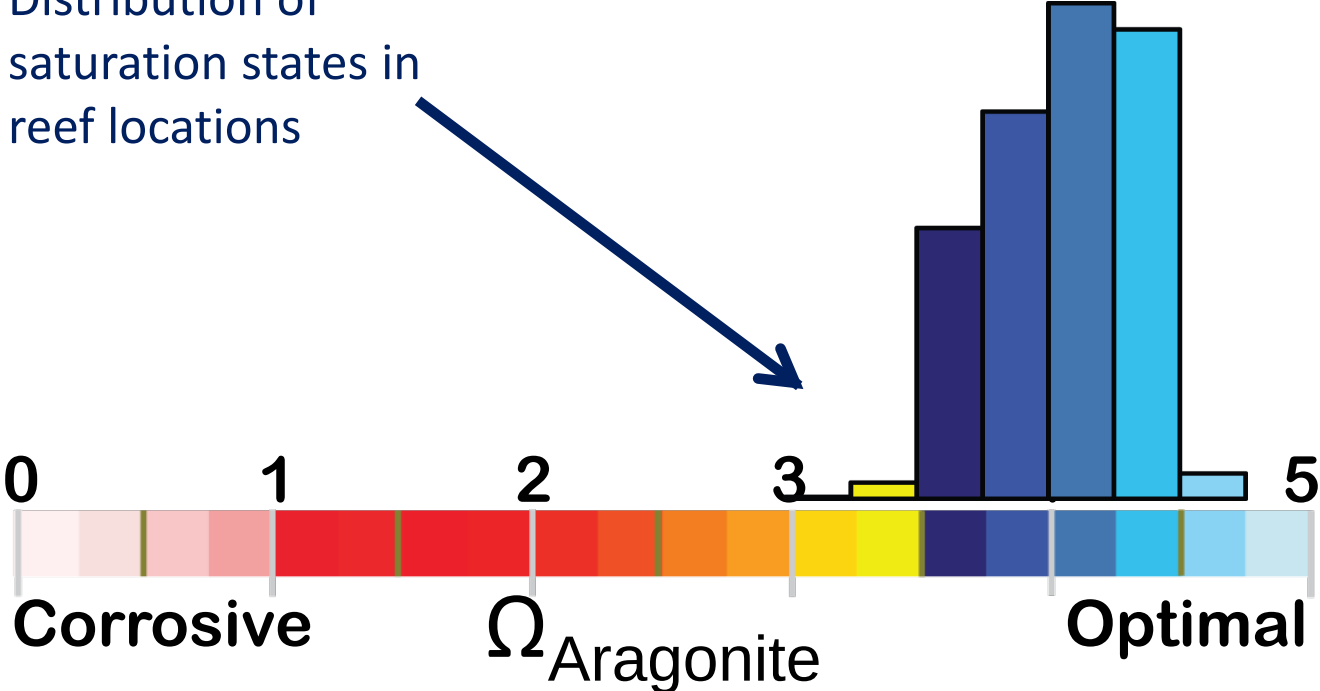


Surface Water
Aragonite
Saturation
States in
Preindustrial
Times

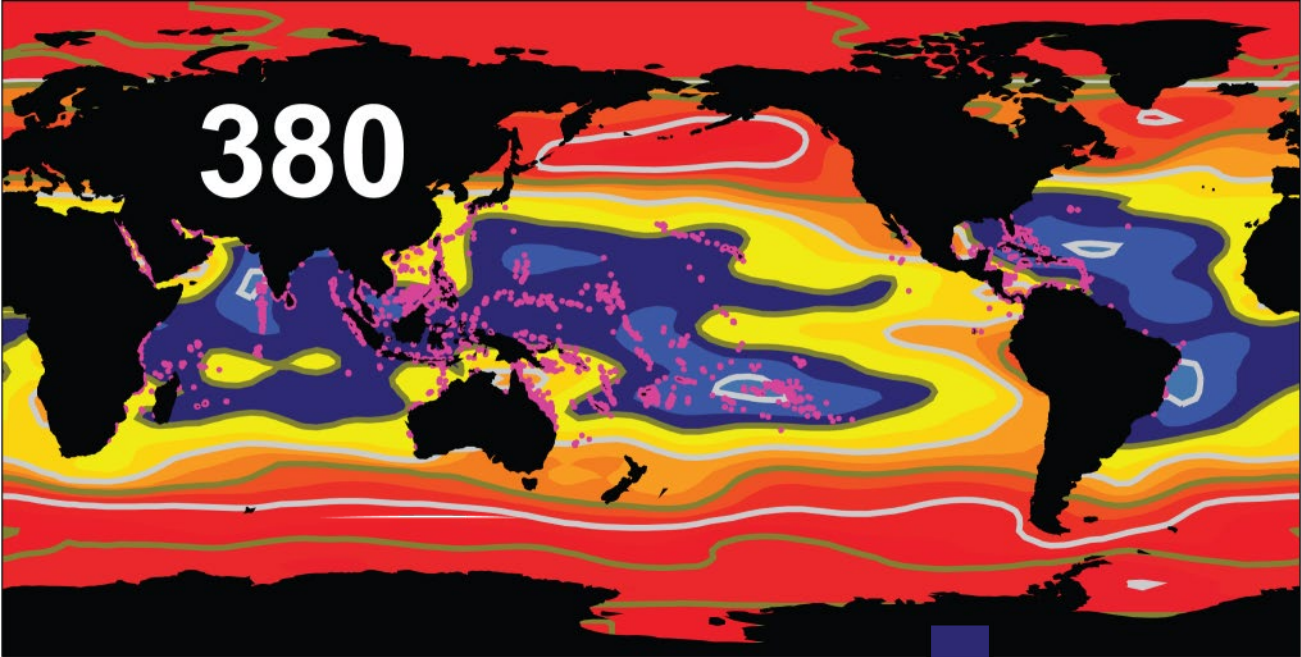


Distribution of
saturation states in
reef locations

Cao and Caldeira, 2008

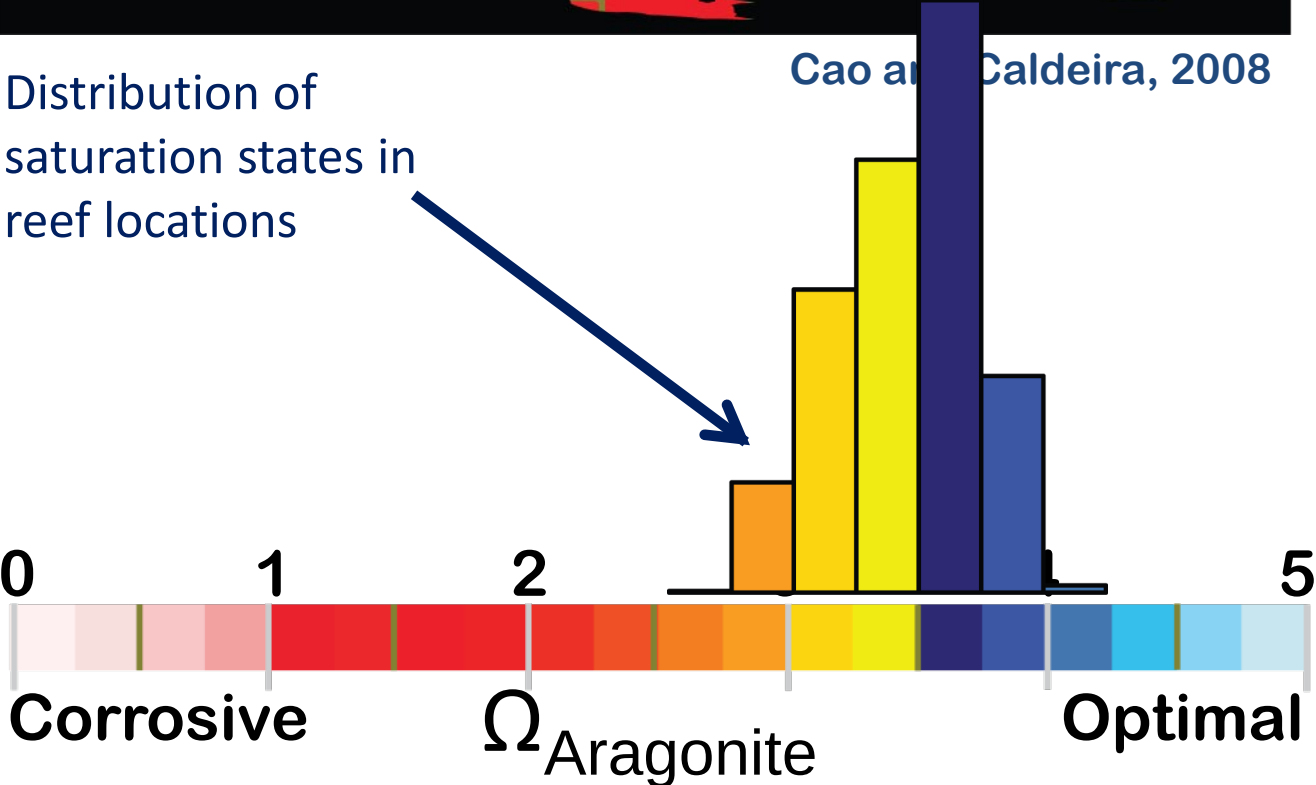


Surface Water
Aragonite
Saturation
States Circa
2000

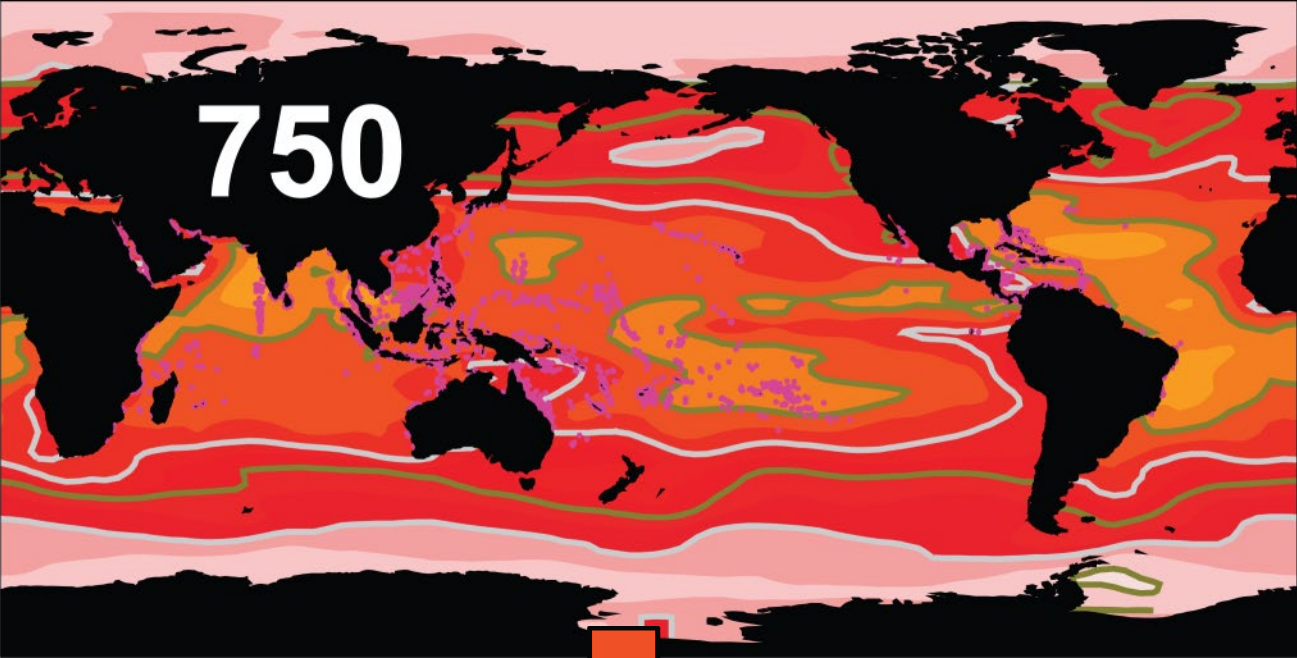


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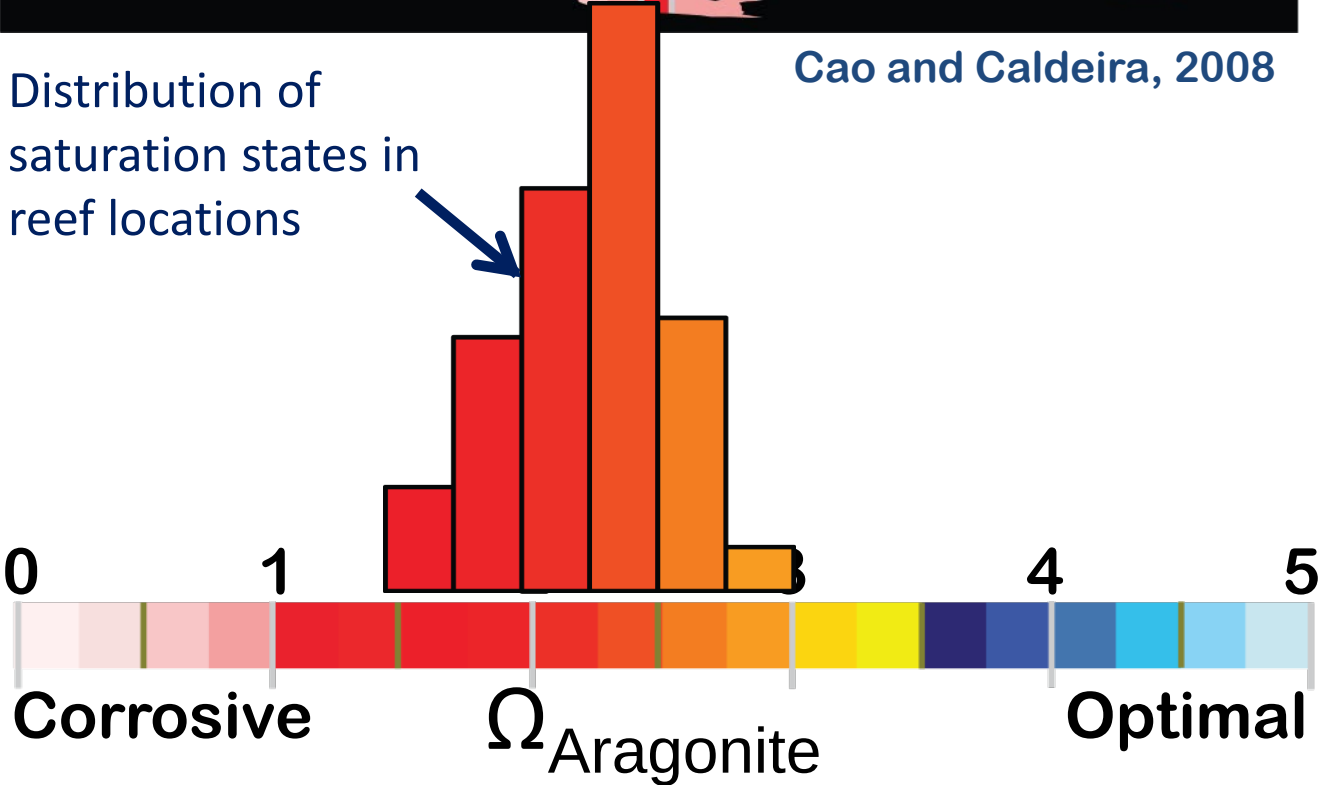


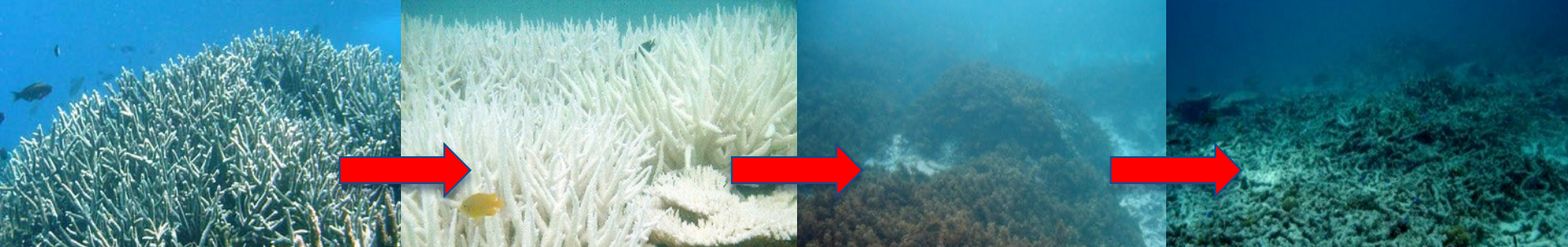
Surface Water
Aragonite
Saturation
States Possible
by 2100 or
Earlier



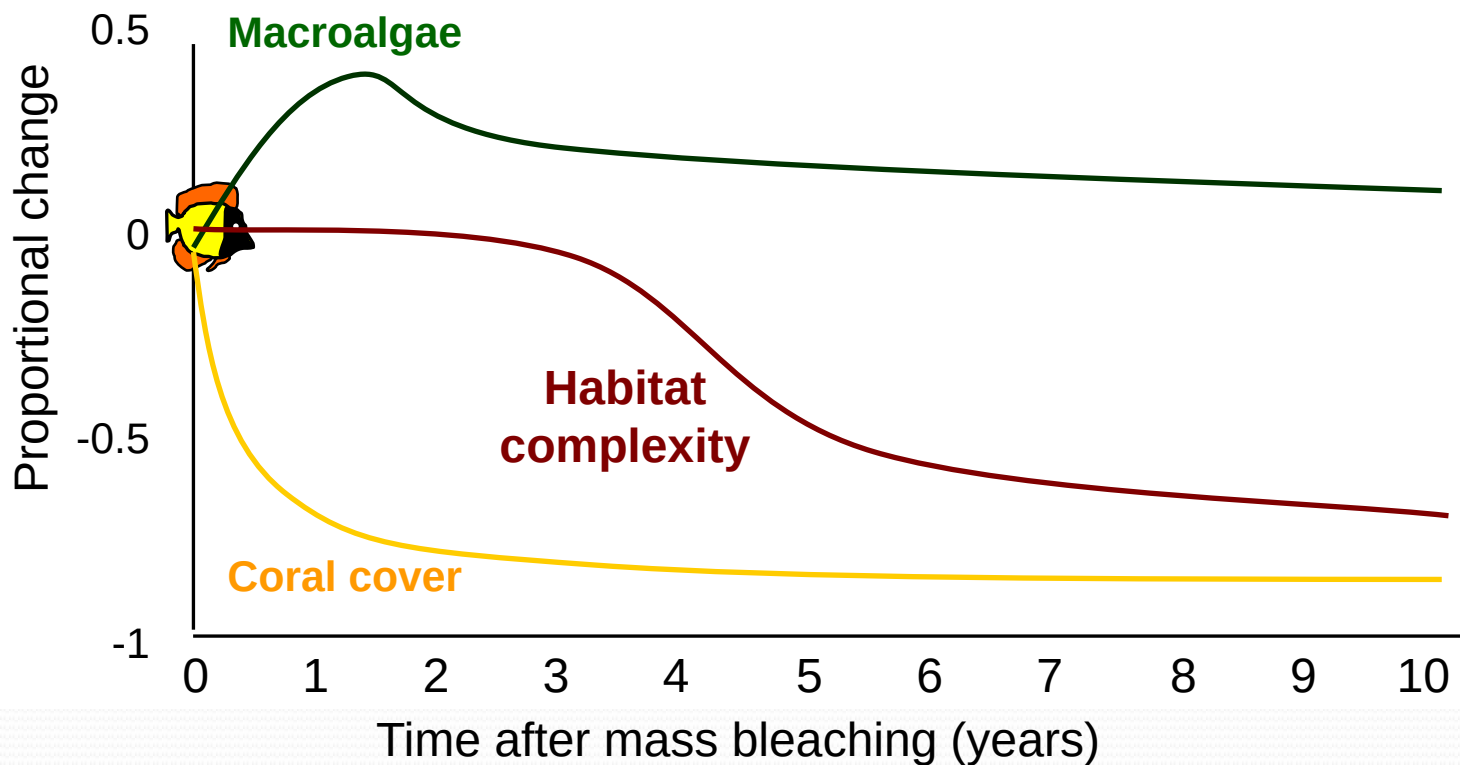
Cao and Caldeira, 2008

Distribution of
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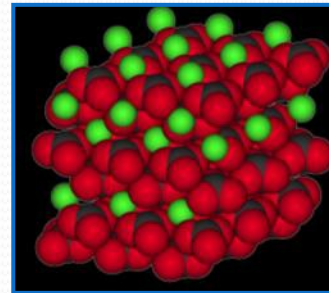
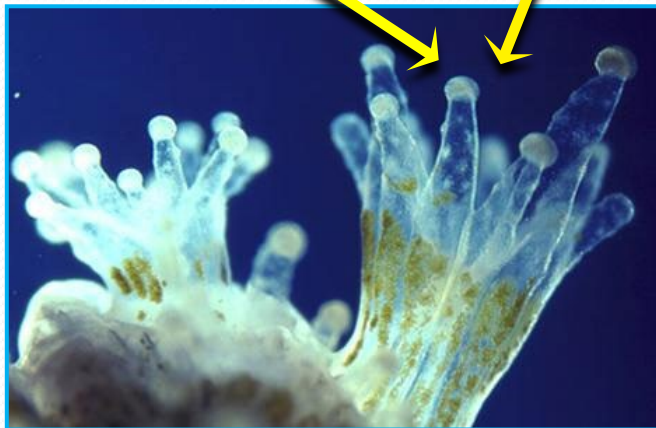
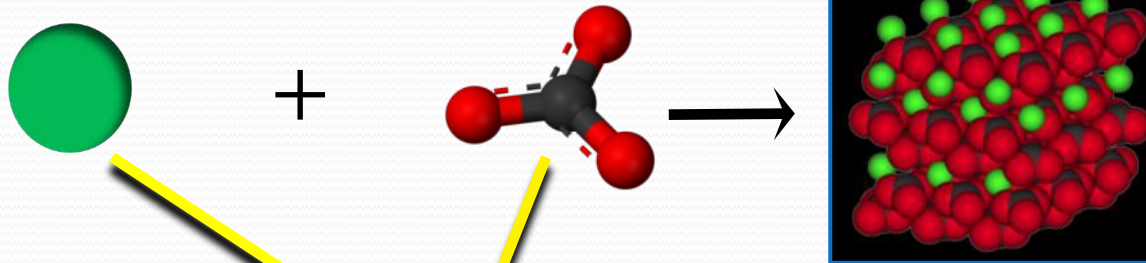
Ocean Acidification will impact coral reef habitat → will indirectly impact reef fish



Small text on the right side of the slide, likely a citation or source information.

Slide courtesy Phil Munday (JCU)

Ocean Acidification: Fundamental Chemistry



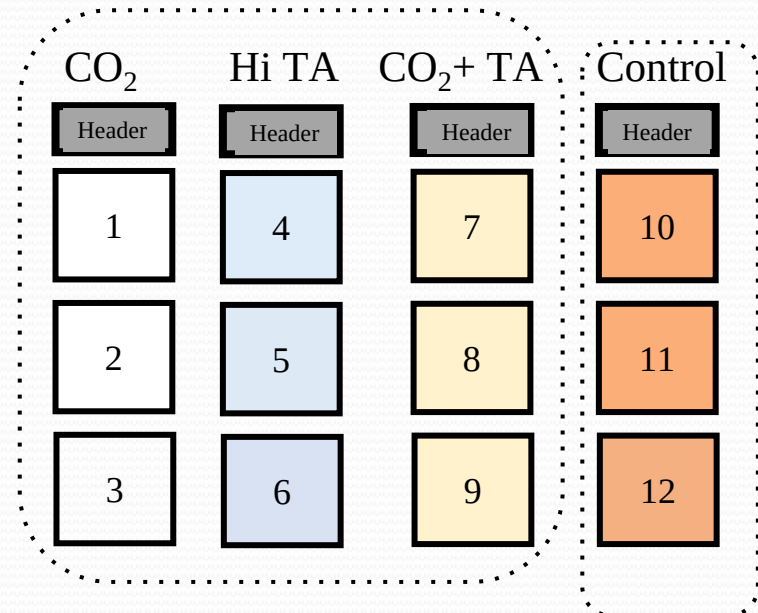
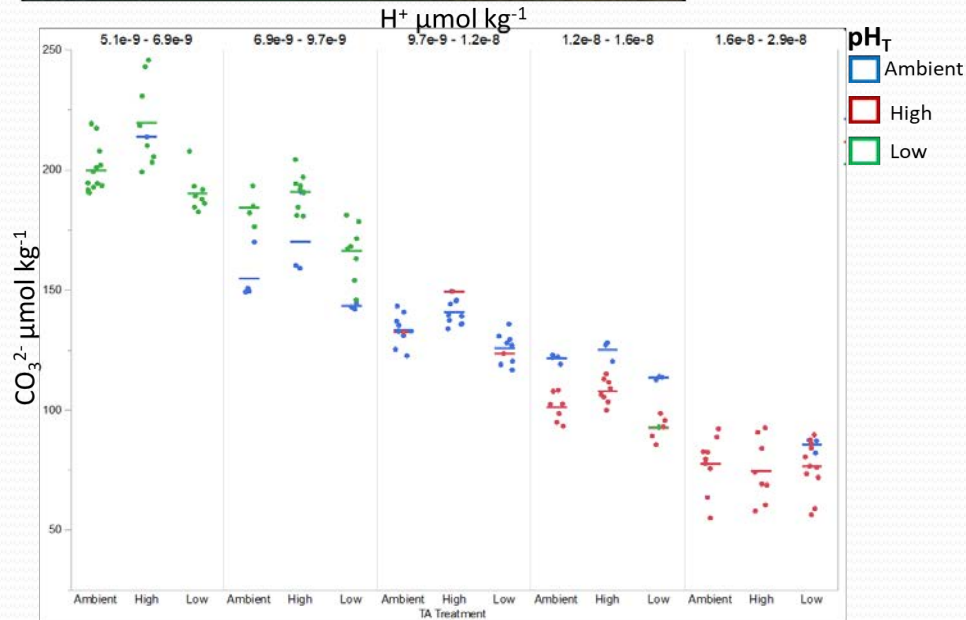
Calcification Index

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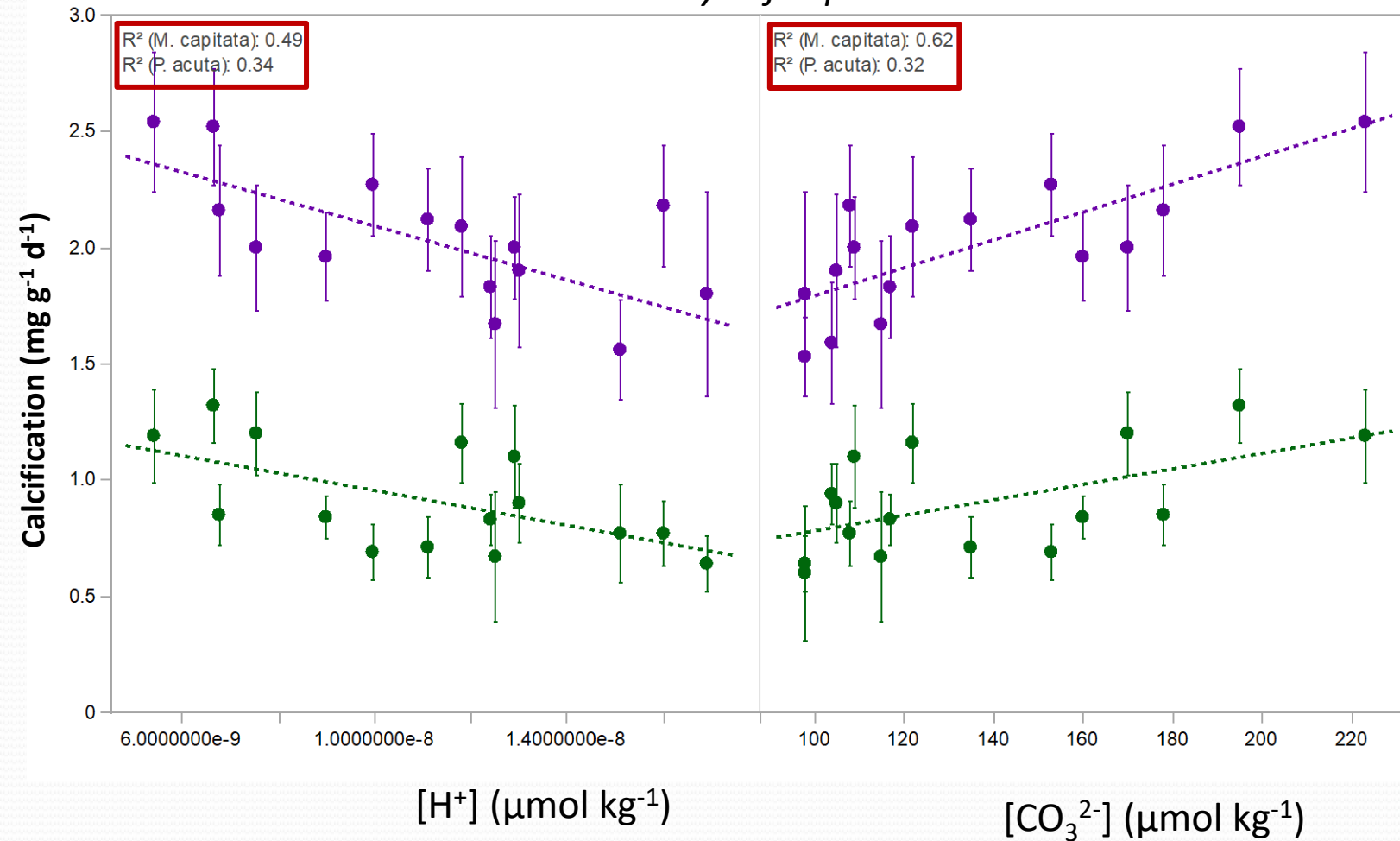


Hawaii Institute of Marine Biology (HIMB)

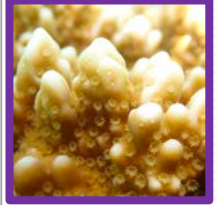


Is Calcification Controlled by H^+ or CO_3^{2-} ?

30 days of exposure



Montipora capitata



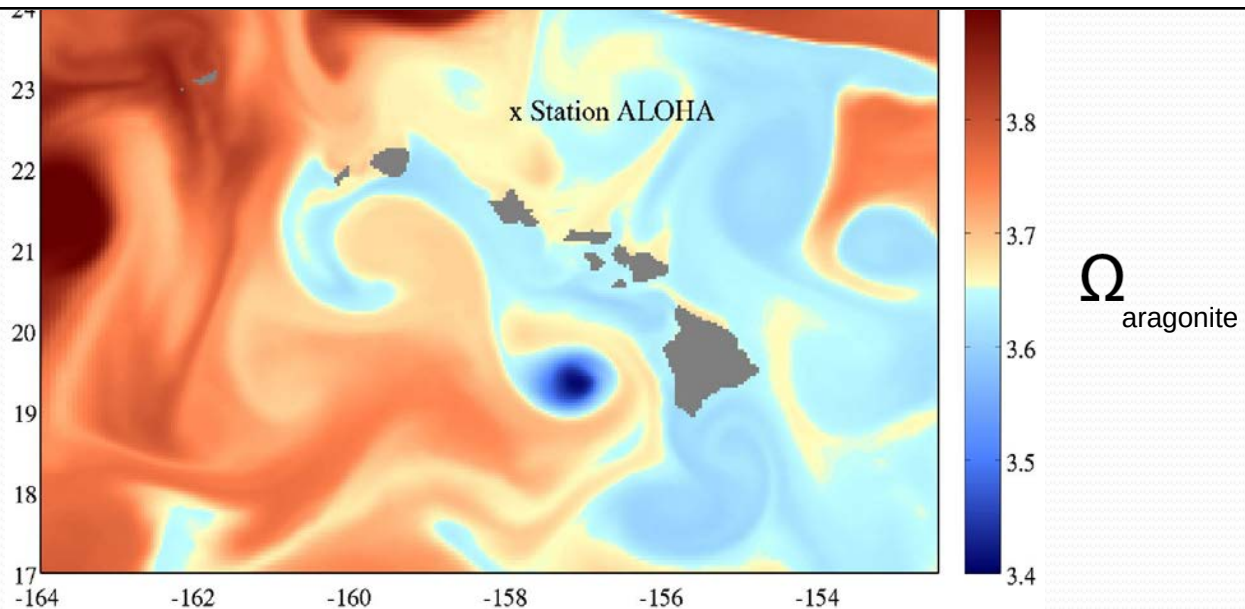
Pocillopora acuta



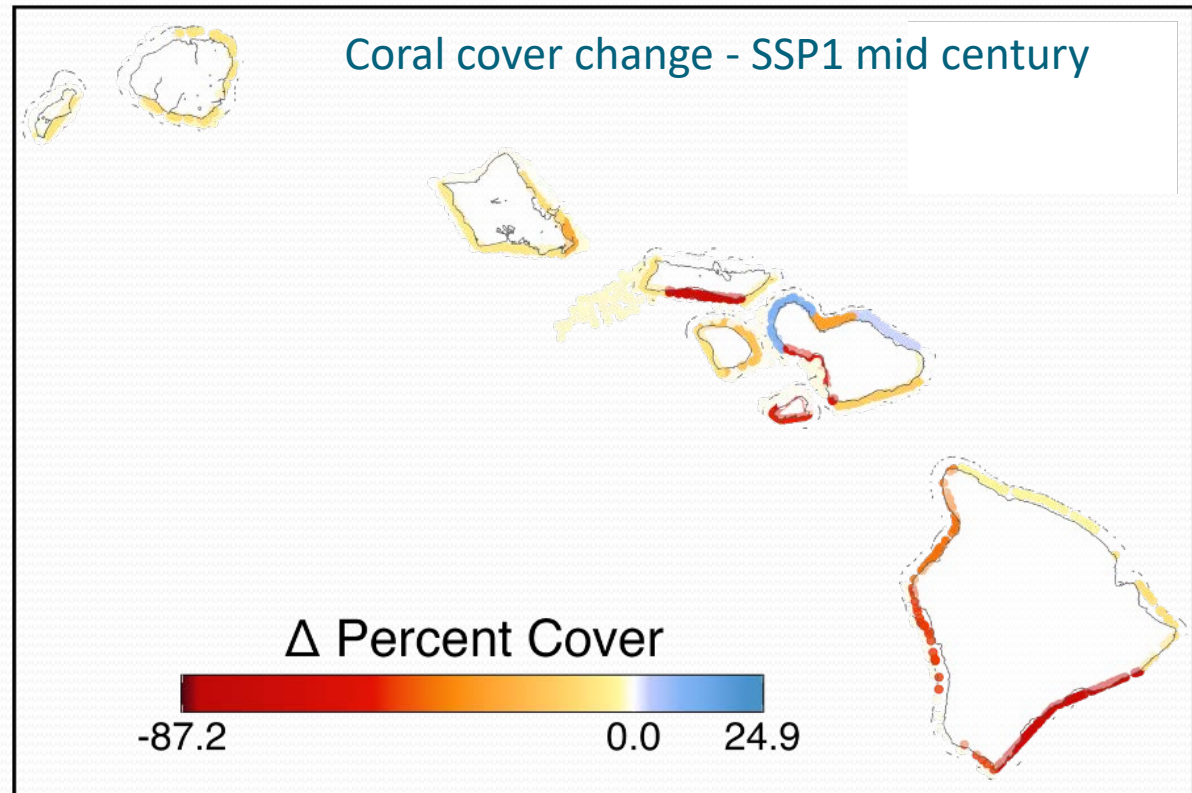
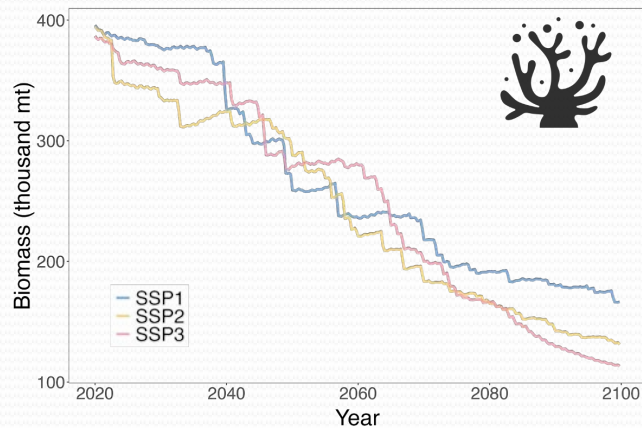
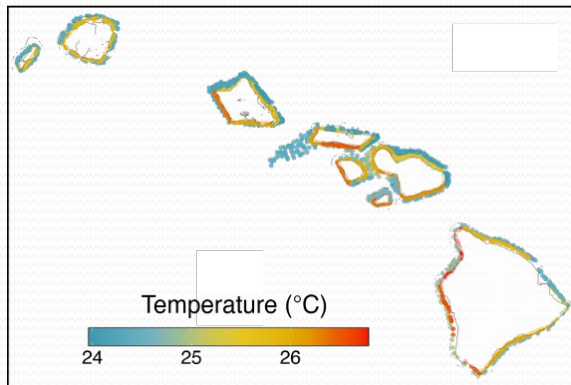
Dynamical Downscaling of Main Hawaiian Islands Domain



*Friedrich et al. 2024:
Submesoscale-permitting
physical/biogeochemical
future simulations for the
main Hawaiian Islands
Liu et al. 2023: Climate
downscaling for regional
models with a neural
network: A Hawaiian
example*



Model-based Changes in Coral Cover from Temperature and OA



Benefit Function Transfer of Recreational Values

Original WTP for sites:

$$\widehat{WTP}_{js} = f(z_{js}, \hat{\beta}_s)$$

Where $j = \text{site}$ & $s = \text{population}$

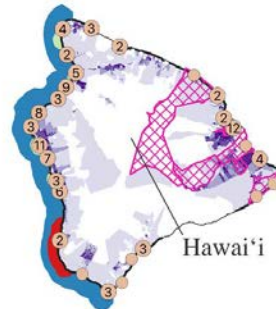
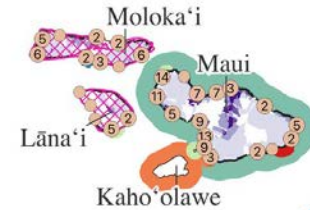
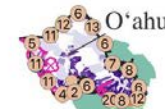
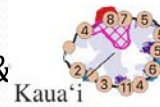
Transferring to *New sites, i, & Population, r*

$$\widehat{WTP}_{ir}^{BT} = f([z_{ir}^1 z_{js}^2], \hat{\beta}_s)$$

where $i \neq j$ & $r \neq s$:

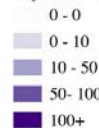
Annual Value :

$$\widehat{WTP}_{ir}^{BT} * \frac{m_r}{m_{s=Maui}} * Pop18_r * 365$$



Identified Disadvantaged Community by Justice40

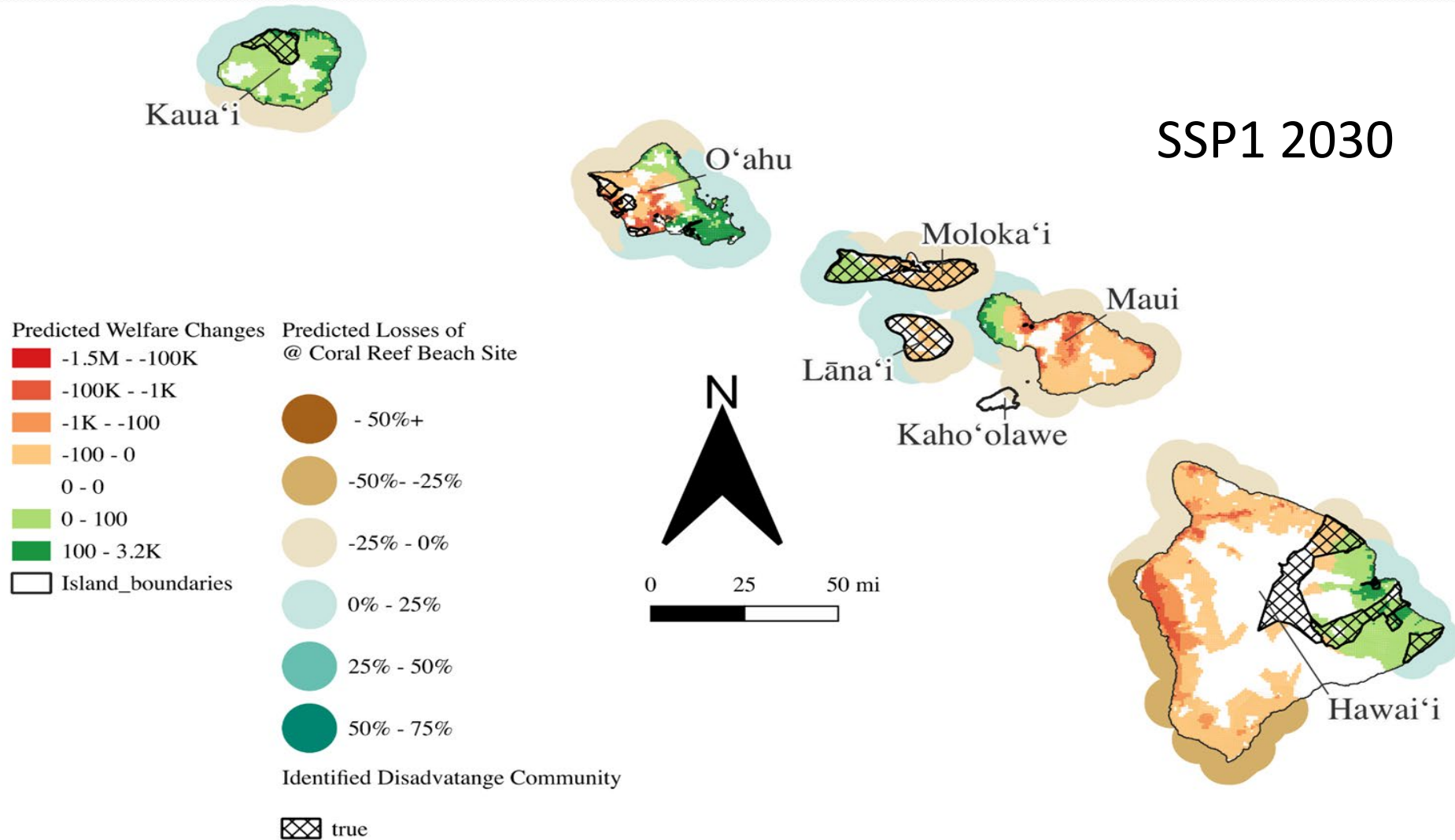
Pop. In Census Blocks



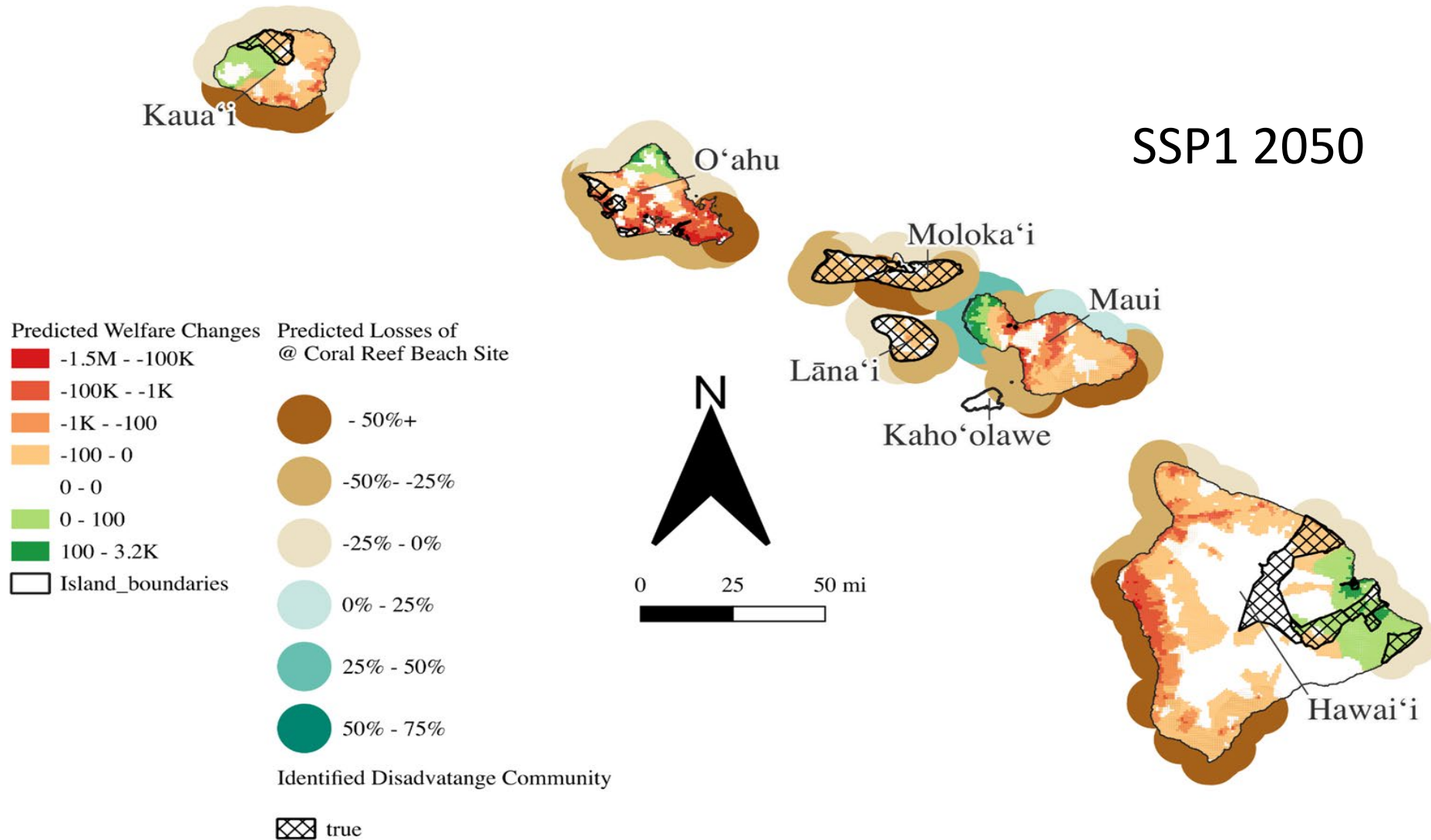
Marine Managed Areas DAR

- Community-Based Subsistence Fishing Area
- Fisheries Management Area
- Marine Life Conservation District
- Lay Net Fishing Prohibited Area
- Fisheries Management Area
- Beach Sites

Impact of Coral Reef Loss on Communities



Impact of Coral Reef Loss on Communities



Impact of Coral Reef Loss on Communities

